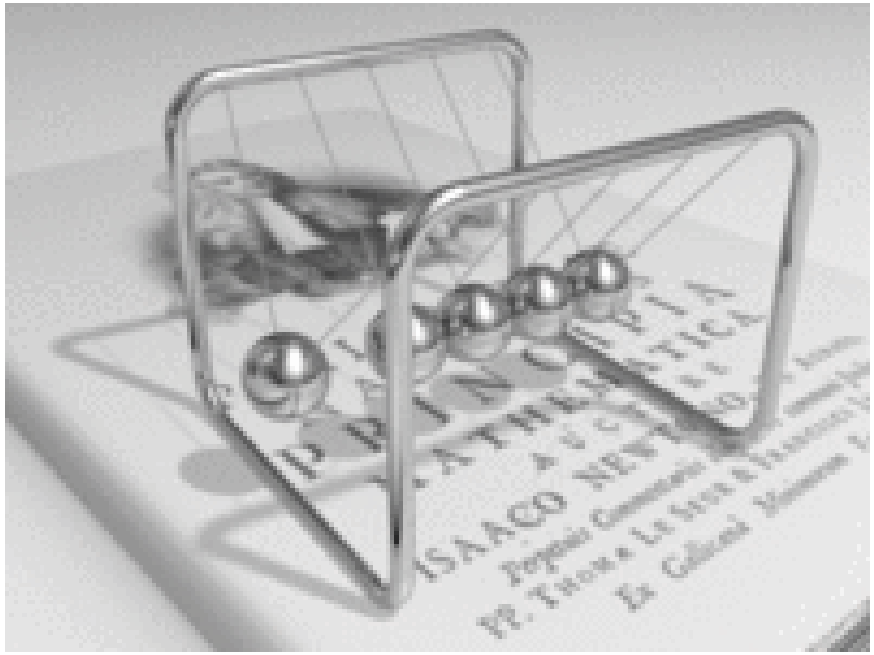
PHYC10009 Foundations of Physics – Lecture 3



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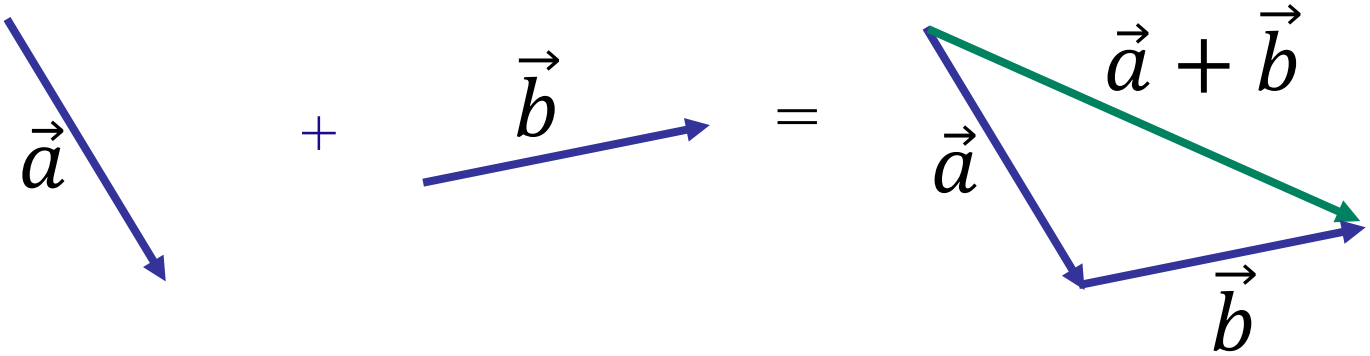
Physics, Room 501

(Please switch off your mobile phone.)

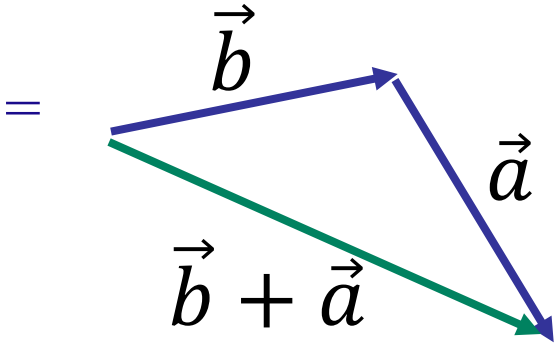
Adding vectors G.2

Place vectors head-to-tail (order doesn't matter)

Result is vector pointing from tail of first to head of second



Note: a vector *only* has magnitude and direction. It is not 'fixed' at a particular point in space. It can be moved around, as long as its magnitude and direction do not change.

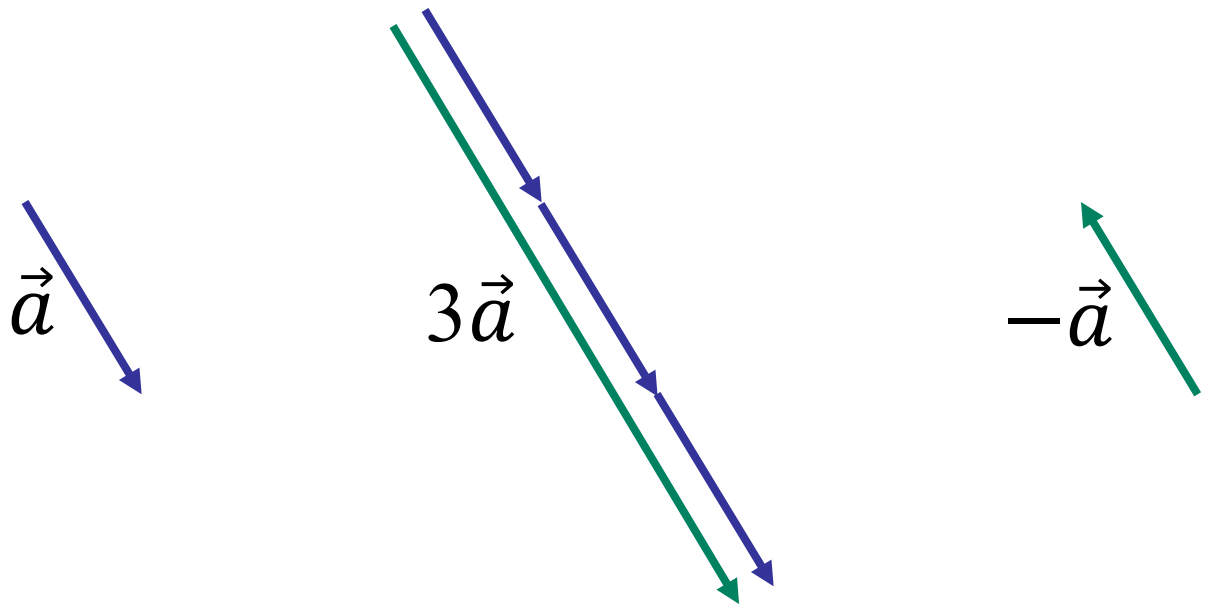


Multiplying a vector by a scalar

Multiply *magnitude* of vector by scalar

Direction unchanged

Multiplication by a negative number reverses direction of vector

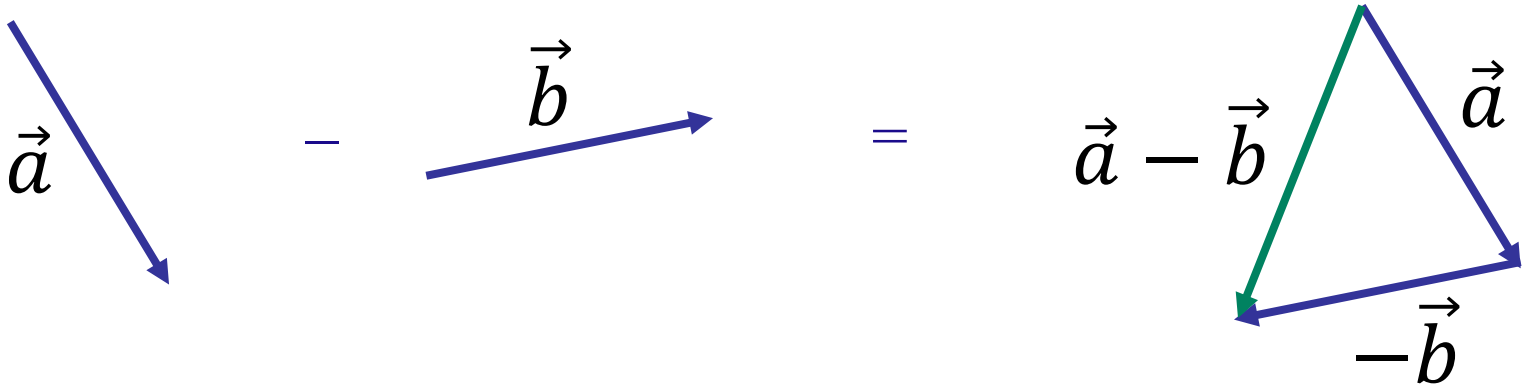


Subtracting vectors - G3.3

Turn second vector around (multiply by –1) and add

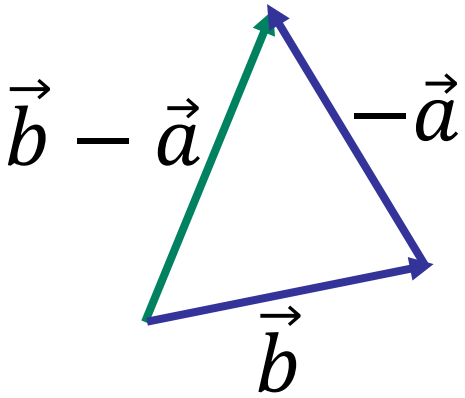
Take care! Don't reverse direction of wrong vector

Order matters, as you can see here:

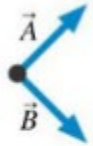


$$\boxed{\vec{a} - \vec{b} \equiv \vec{a} + (-\overrightarrow{b})}$$

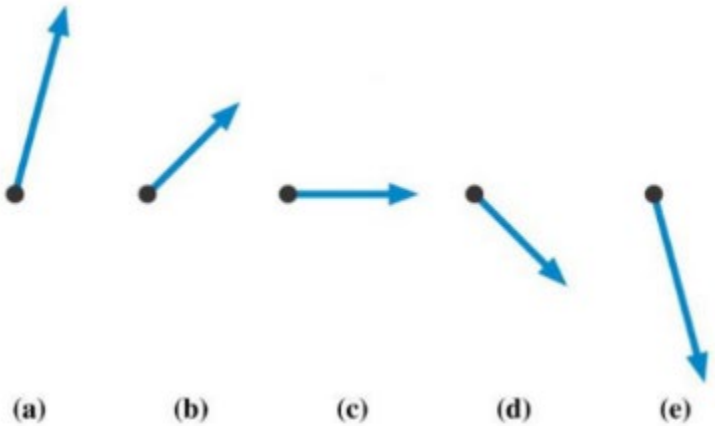
$$\vec{b} - \vec{a} \quad = -(\vec{a} - \vec{b})$$



Vector Addition - 2

Given 

which figure shows $2\vec{A} - \vec{B}$?



(a)

(b)

(c)

(d)

(e)

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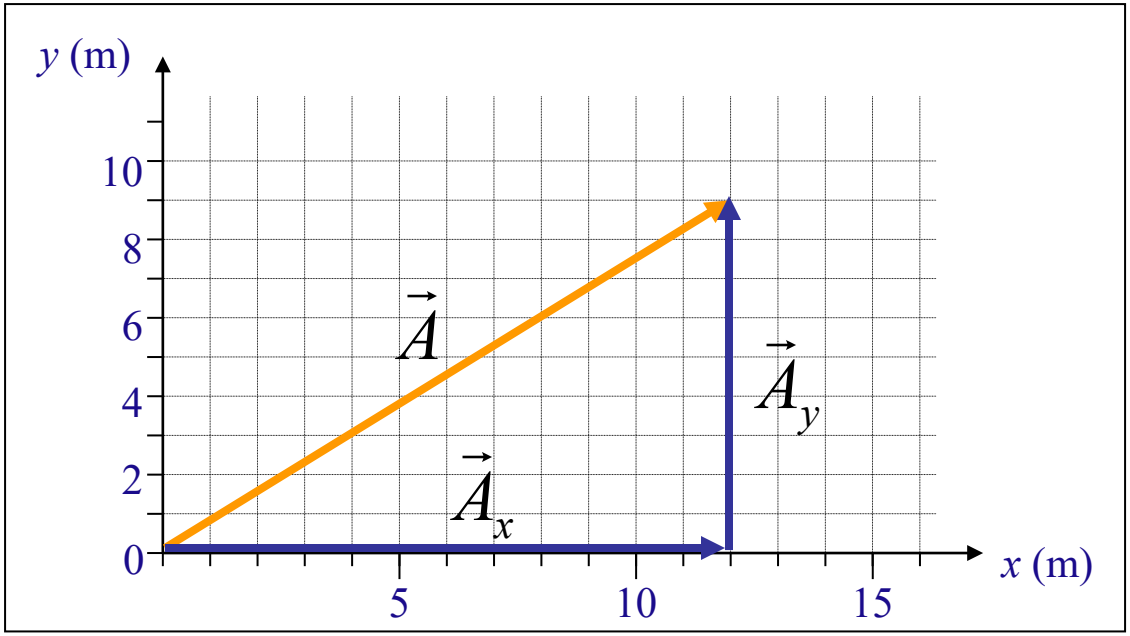
Components of vectors - G 3.4



$$\vec{A} = \vec{A}_x + \vec{A}_y$$

Separate vectors into *orthogonal components*

"Orthogonal" means at right angles

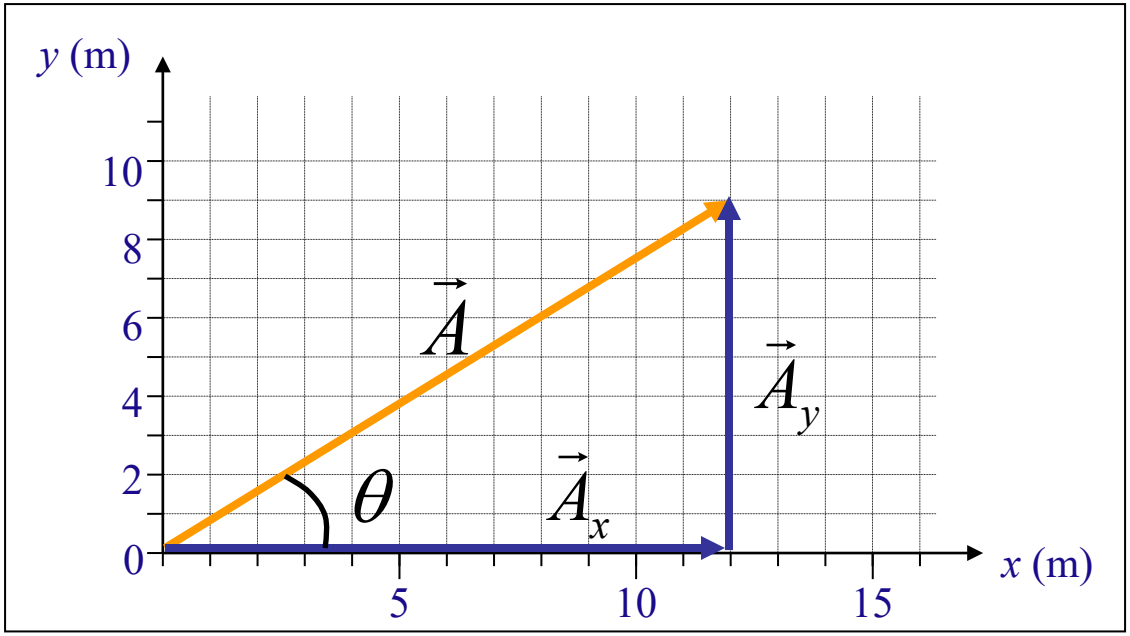
Full vector = sum of two orthogonal components

Why?

Easy to add/subtract arbitrary vectors (just add/subtract components)

Physics acts along one direction only (e.g. gravity)

To find the components of a vector - G3.4



Given: $|\vec{A}| = 15\text{ m} \qquad \theta = 36.7°$

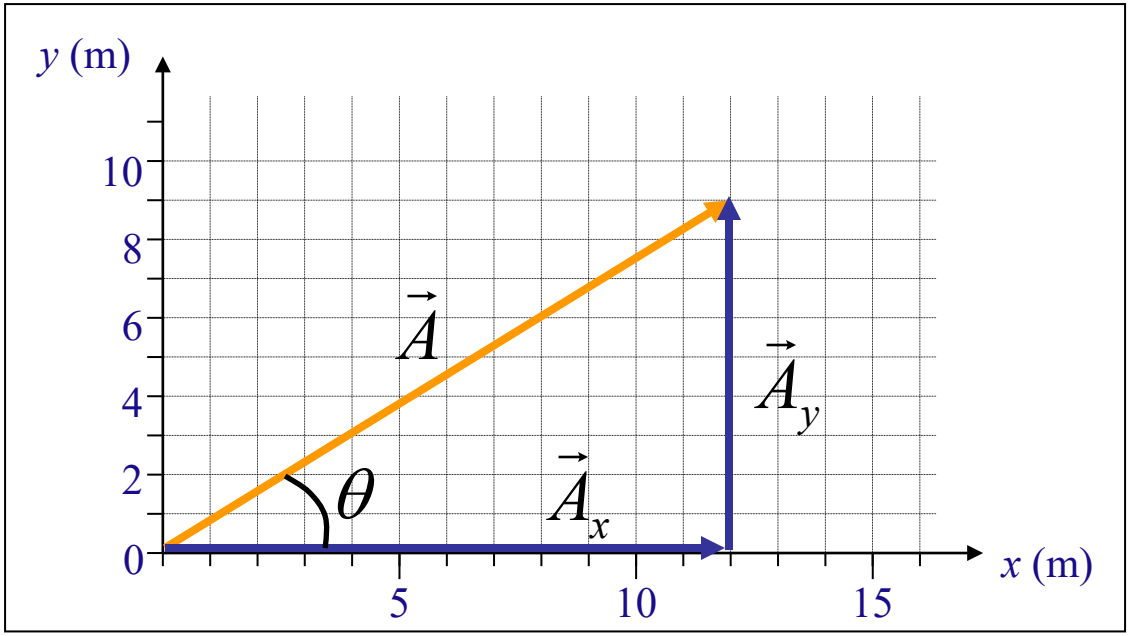
We find: $A_x = 15\cos 36.7° = 12\text{ m}$

$A_y = 15\sin 36.7° = 9\text{ m}$

$$A_x = A\cos\theta$$

$$A_y = A\sin\theta$$

To find a vector from its components – G3.4



Given: $A_x = 12\text{ m}$ $\quad A_y = 9\text{ m}$

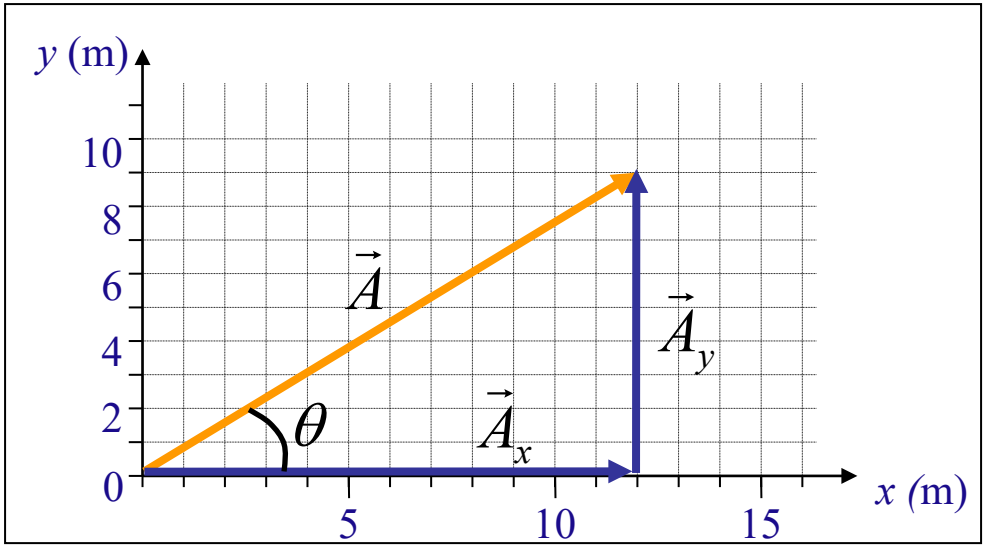
We find: $A = |\vec{A}| = \sqrt{12^2 + 9^2} = 15\text{ m}$

$$\tan\theta = \left|\frac{9}{12}\right| \Rightarrow \theta = 36.7°$$

$$A = |\vec{A}| = \sqrt{A_x^{\,2} + A_y^{\,2}}$$

$$\tan\theta = \left|\frac{A_y}{A_x}\right|$$

Notation



Original vector: $\vec{A} = \left(A_x, A_y\right)$

$= (12, 9)$

Magnitude of original vector: $A = |\vec{A}|$

Component *vectors*: $\vec{A}_x = (12,0) = 12\hat{x}$

$\vec{A}_y = (0,9) = 9\hat{y}$

Components (scalars):

$$A_x = |\vec{A}_x| \qquad A_y = |\vec{A}_y|$$

Note: If component vectors are in the negative direction, must put in signs of components specifically!

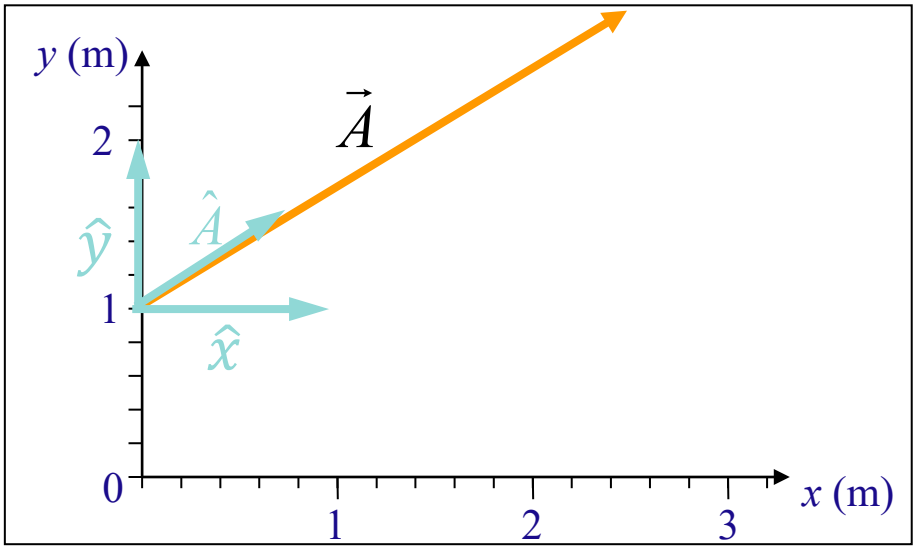
Unit vectors

A *unit vector* is any vector that has magnitude = 1

Notation: add "hat":

Create your own unit vector: $\hat{A} = \dfrac{\vec{A}}{|\vec{A}|}$ Unit vector in direction of $\vec{A}$

Unit vectors most common along coordinate axes: $\hat{x}$, $\hat{y}$



- Can now write arbitrary vector in terms of unit vectors:

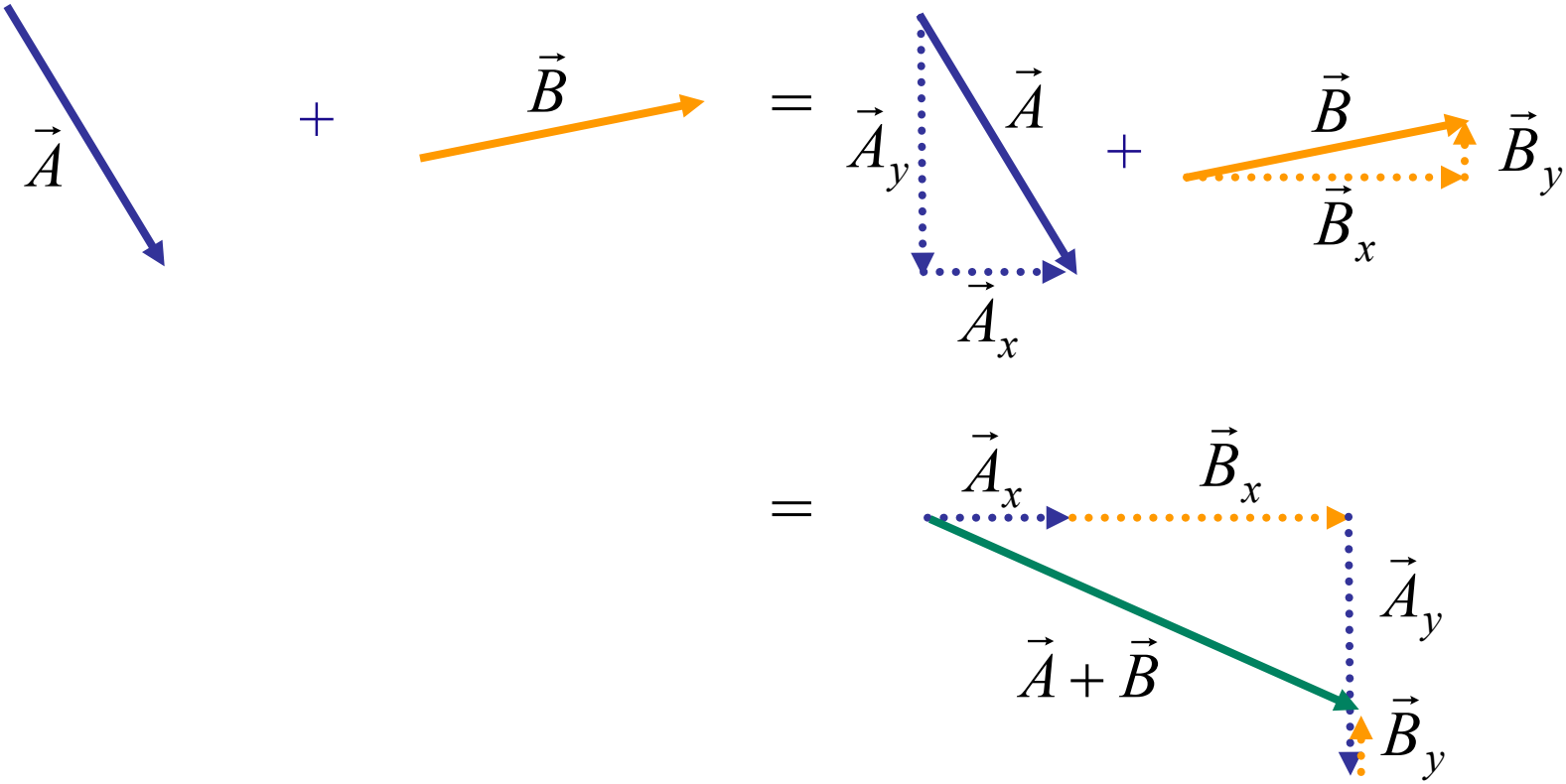
$$\vec{A} = \vec{A}_x + \vec{A}_y = (A_x, A_y)$$
$$= A_x\hat{x} + A_y\hat{y}$$

Adding vectors using components – G3.4

Step 1: Find the components of each of the vectors

Step 2: Add x and y components separately

Step 3: Find the resultant vector from its components



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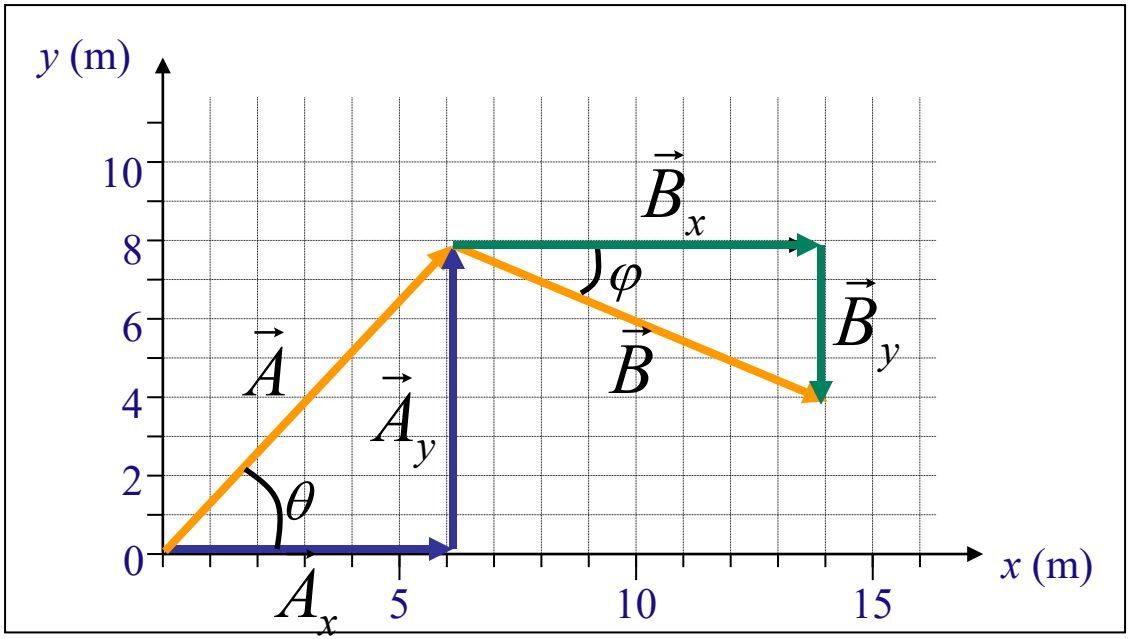


$$A_x = A\cos\theta = 6\text{m} \qquad A_y = A\sin\theta = 8\text{m}$$

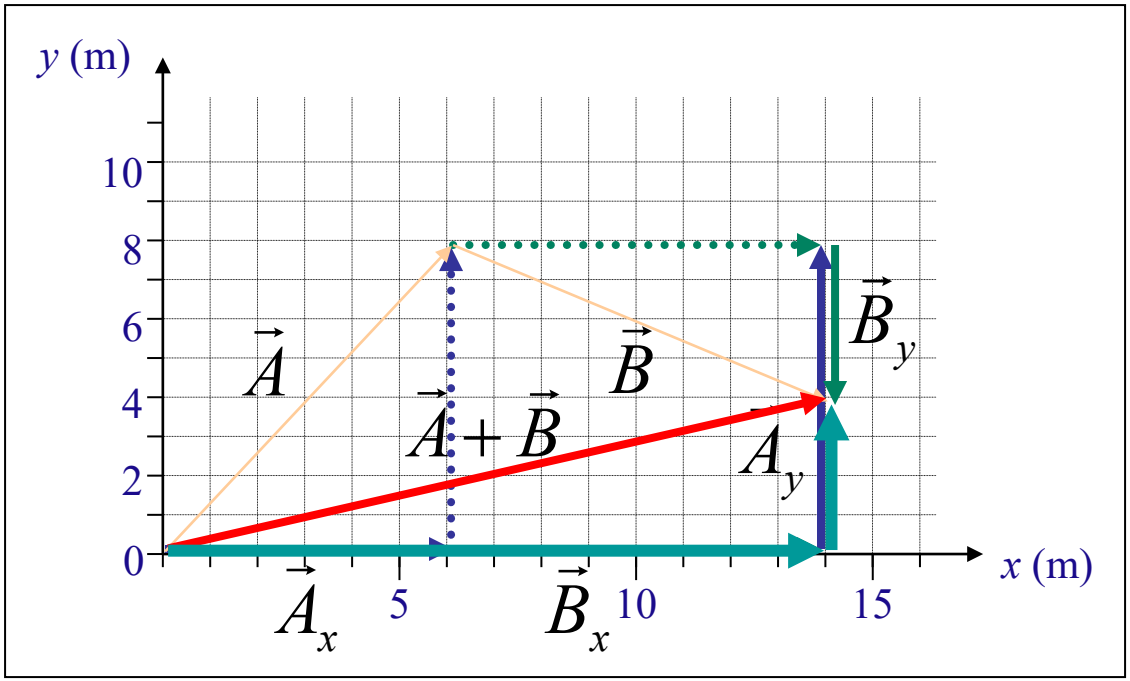
$$B_x = B\cos\varphi = 8\text{m} \qquad B_y = B\sin\varphi = -4\text{m}$$

Adding vectors using components - G 3.4

Step 1: Find the components of each of the vectors

Step 2: Add x and y components separately

Step 3: Find the resultant vector from its components



$$(A_x, A_y) = (6\text{m}, 8\text{m})$$

$$(B_x, B_y) = (8\text{m}, 4\text{m})$$

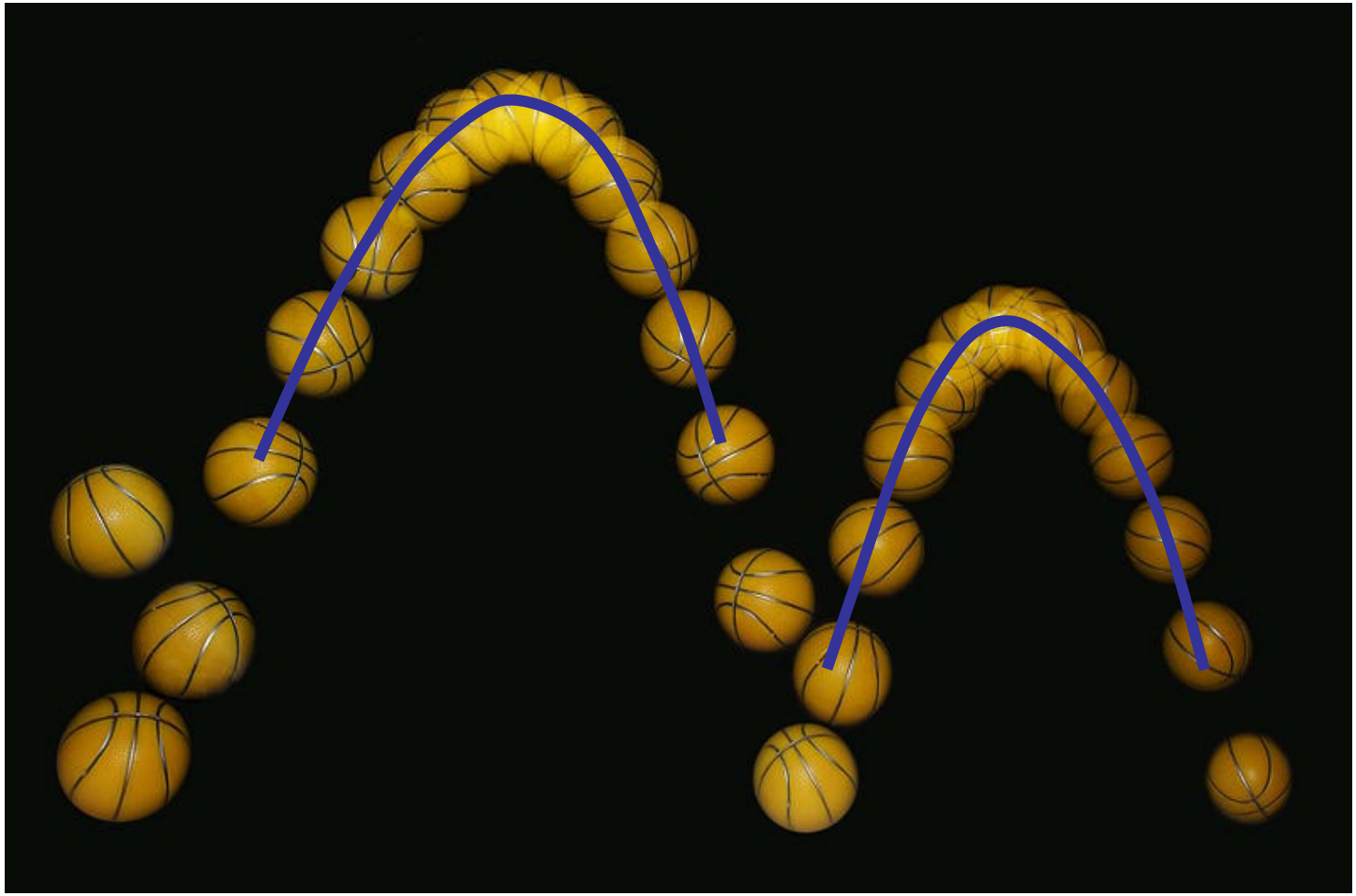
$$A_x + B_x = 14, A_y + B_y = 4$$

$$\vec{A} = (14m, 4m) = 14\hat{x} + 4\hat{y}$$

$$|\vec{A} + \vec{B}| = \sqrt{14^2 + 4^2} = 14.6\text{ m}$$

$$\tan\alpha = \frac{4}{14}$$

Motion in 2d: projectile motion - G 3.5



Projectile motion - G3.5

- A *projectile* is any object that is thrown, projected or dropped, thereafter moving under the dominant influence of gravity.

Kinematics of projectiles

- Horizontal and vertical motions are *independent* (i.e. one does not affect the other).
- If gravity is the only force acting (no air resistance):
  - Vertical acceleration is constant and equal to the free-fall acceleration due to gravity: downwards with magnitude $g = 9.8\ ms^{-2}$.
  - Horizontal acceleration is *zero*.

D Ballistic Cart

Projectile motion - G 3.5

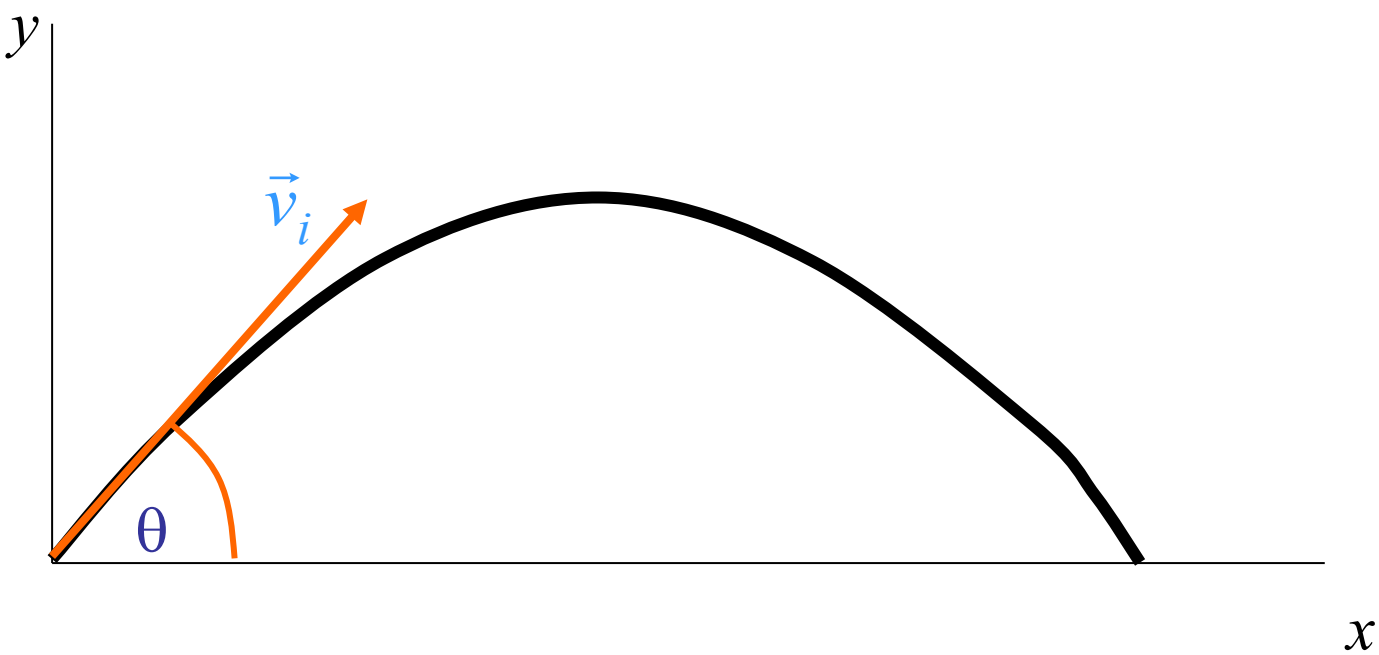
Projectile $\Rightarrow$ object is moving under influence of only gravity

Launched with velocity $\vec{v}_i$

Speed v_i

Launch angle θ (with respect to horizontal)

Parabolic trajectory



Projectile motion - G3.5

Object moving under influence of only gravity

Parabolic trajectory

Note "*x*" is generic displacement
Can be in direction *x* or *y*

Recall $x_f = x_i + v_i \Delta t + \frac{1}{2} a (\Delta t)^2$

Horizontal component of acceleration is *zero*

$a_x = 0$

$$x_f = x_i + v_{ix} \Delta t$$

$$v_{fx} = v_{ix}$$

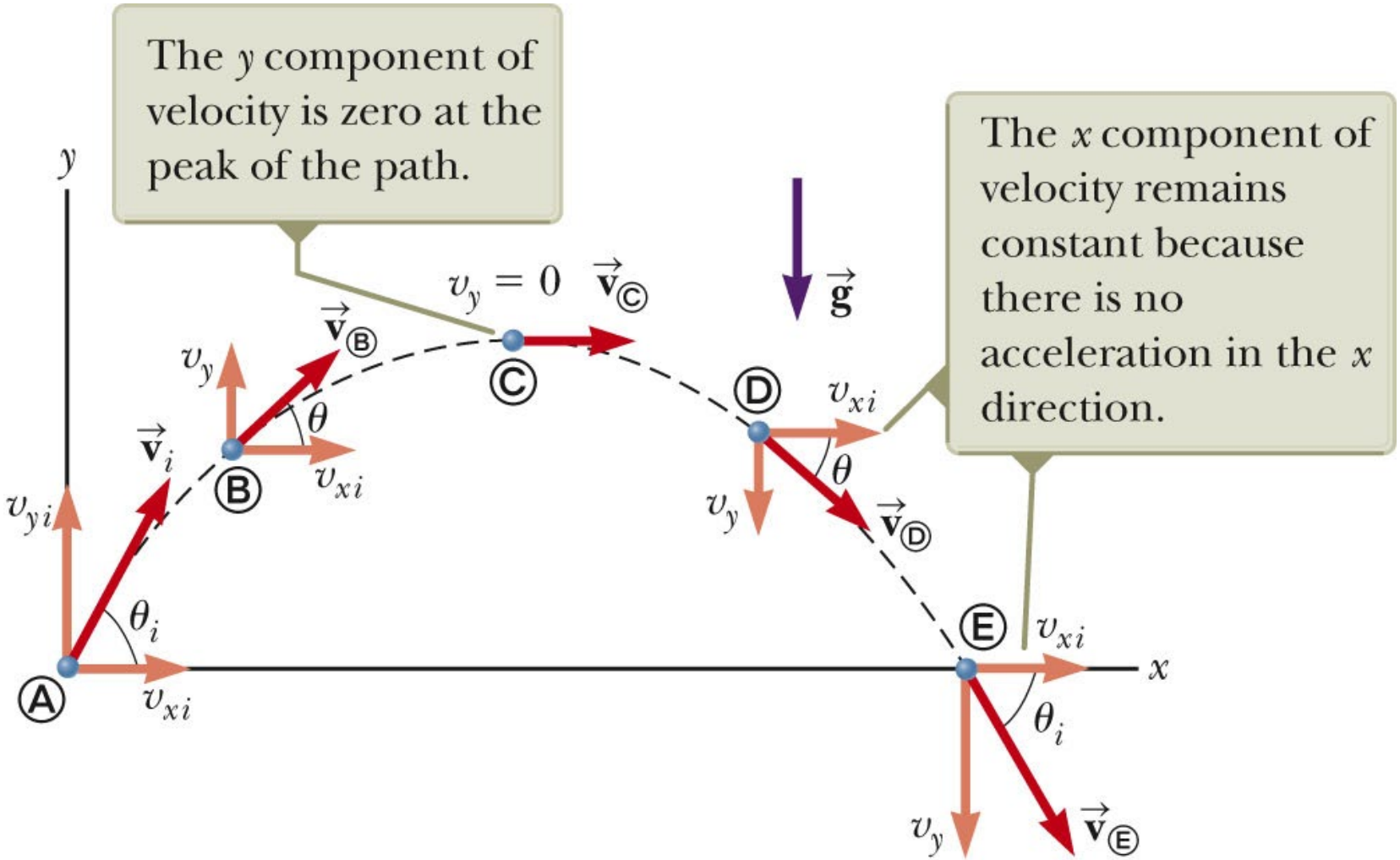
Vertical component of acceleration is -g

$a_y = -g$

$$y_f = y_i + v_{yi} \Delta t - \frac{1}{2} g (\Delta t)^2$$

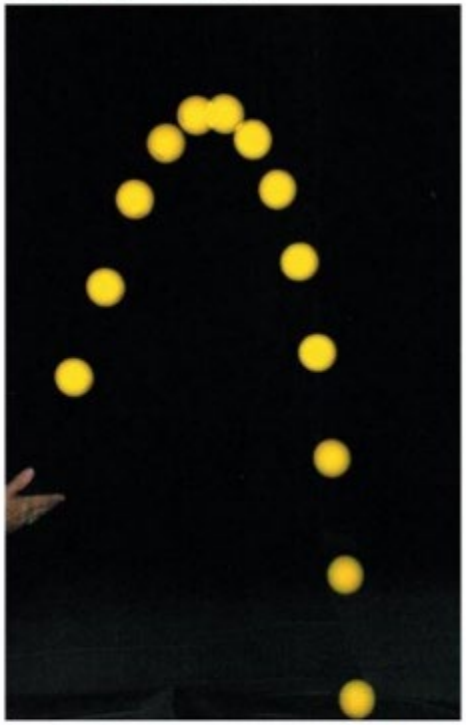
$$v_{yf} = v_{yi} - g(\Delta t)^2$$

Projectile motion - G 3.5



Tossed Ball

At the highest point of the ball's flight, the ball is …



accelerating upwards

accelerating downwards

accelerating to the right

not accelerating (a=0)

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Equations of projectile motion - G 3.5

$$v_f = v_i + a\Delta t$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a (\Delta t)^2$$

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$$

$$2a\Delta x = v_f^{\,2} - v_i^{\,2}$$

Horizontal motion (x direction):

$$a_x = 0$$

$$(v_x)_f = (v_x)_i$$

$$x_f = x_i + v_i \Delta t$$

Vertical motion (y direction):

$$a_y = -g$$

$$(v_y)_f = (v_y)_i - g\Delta t$$

$$y_f = y_i + (v_y)_i \Delta t - \frac{1}{2} g (\Delta t)^2$$

No need to remember these equations separately. Just remember the constant-acceleration equations!

Projectile motion problem solving - G 3.6

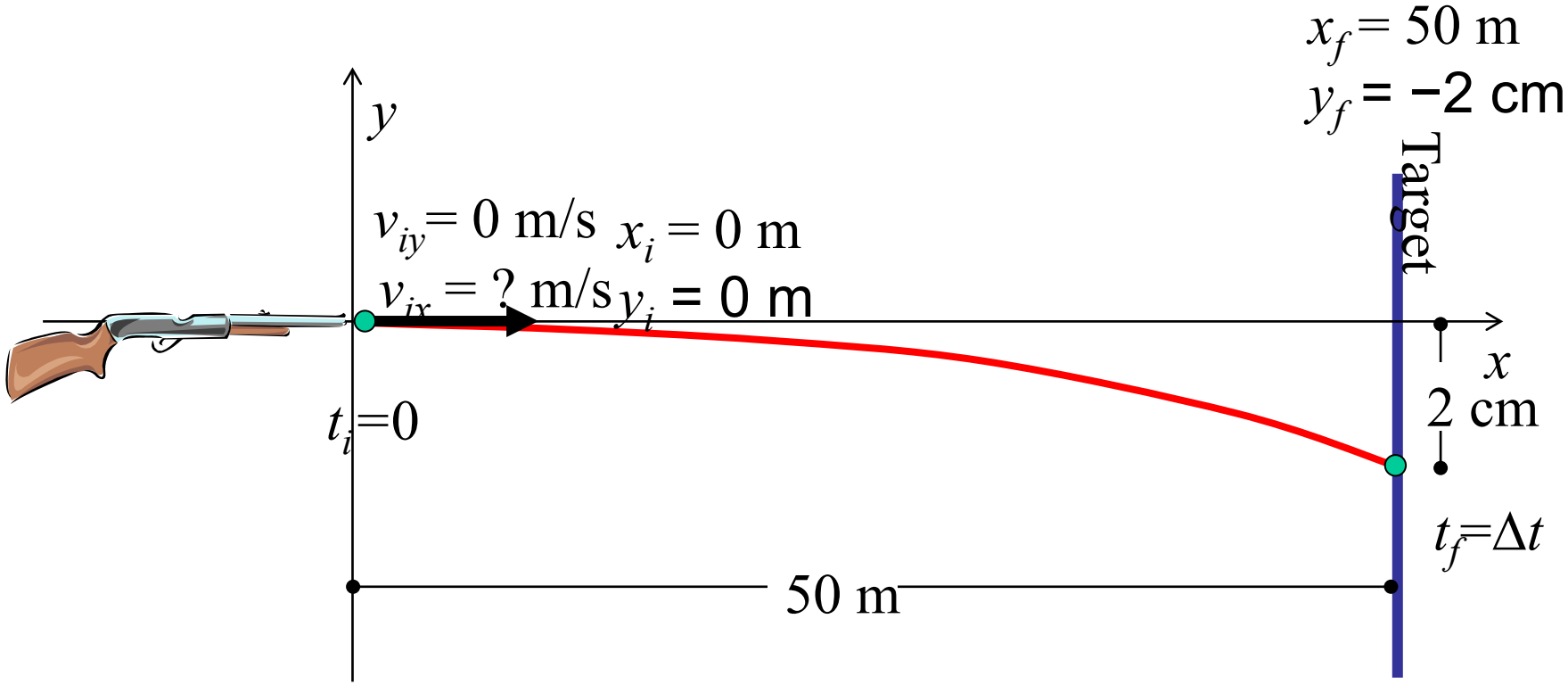
A rifle is aimed horizontally, i.e. with barrel parallel to the ground, at a target 50m away. The bullet hits the target 2cm below the aim point, i.e. below the height of the gun.

i. What was the bullet flight time?
ii. What was the bullet speed when it left the barrel of the rifle?

Assume bullet is point particle
Assume free-fall motion
No air-resistance

3.7 Projectile motion problem solving

Visualise:



Find: (i) Time elapsed Δt; (ii) v_{ix}

Projectile motion problem solving - G3.6

Solve: $y_f = \cancelto{0}{y_i} + \cancelto{0}{v_{iy}}\Delta t - \frac{1}{2}g(\Delta t)^2$

$$\Delta t^2 = \frac{-2y_f}{g} \Rightarrow \Delta t = \sqrt{\frac{-2\times(-0.02)}{9.8}} = 0.06 \text{ s}$$

$$x_f = \cancelto{0}{x_i} + v_{ix}\Delta t$$

$$v_{ix} = \frac{x_f}{\Delta t} = \frac{50}{0.06} = 800\ ms^{-1}$$

(i) $\Delta t = 0.06$ s

(ii) Speed $= 8 \times 10^2$ m/s

Projectile motion problem solving - G3.6

Assess:

Have you answered the question? ✓

Are units, signs and significant figures correct?

Units (i) time (s) and (ii) speed (m/s)

No sign issues

Data given with 1-2 significant figures ✓

Answer given with 1 sig fig

Vector answer where appropriate! ✓

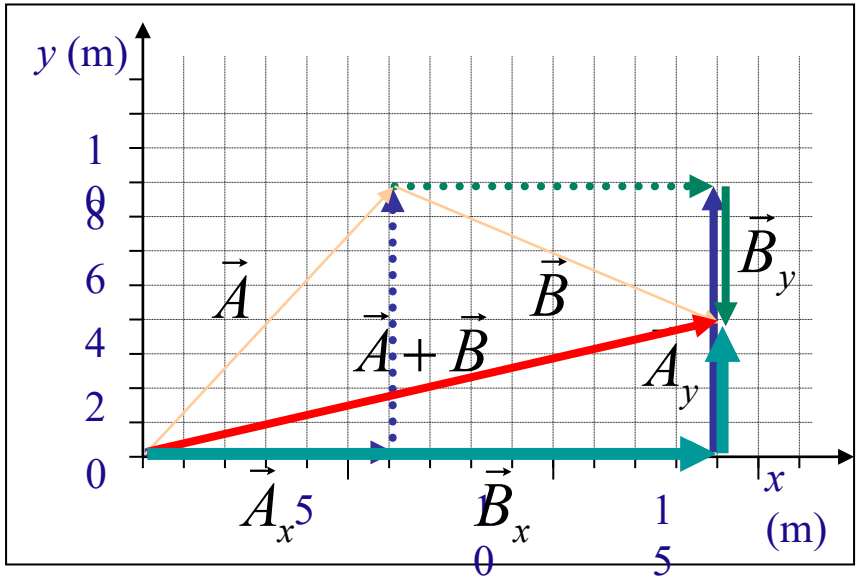
Only speed and time requested, so no vector needed

Is answer reasonable? ✓

Yes. Would expect short time and fast speed since the bullet has dropped only 2cm in the time it takes to travel 50 m horizontally

Summary of Lecture 3

Vector components:
Separately algebraically sum the x and y components when adding vectors



Horizontal motion (x direction):

$$a_x = 0$$

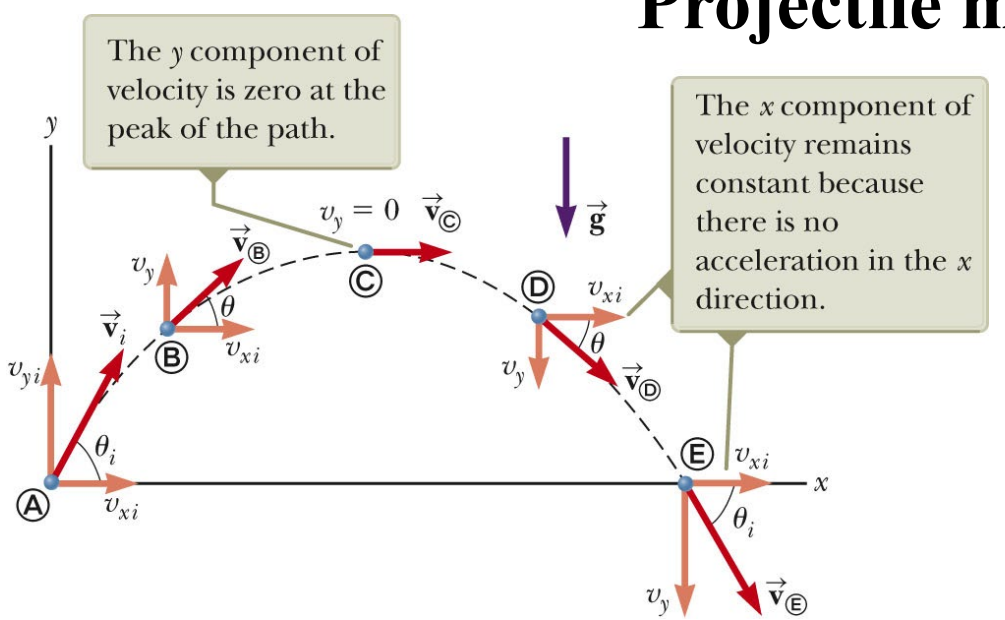
$$(v_x)_f = (v_x)_i$$

$$x_f = x_i + v_i \Delta t$$

Vertical motion (y direction):

$$a_y = -g$$

$$(v_y)_f = (v_y)_i - g\Delta t$$

$$y_f = y_i + (v_y)_i \Delta t - \frac{1}{2} g (\Delta t)^2$$

Projectile motion